SIDDHARTH

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SUMMARY

Postgraduate in Chemistry from IIT Roorkee with a strong research foundation in computational materials science, thermoelectrics, perovskite and catalysis. Experienced in DFT (VASP, Gaussian), Boltzmann transport theory (BoltzTraP2) and MD simulations (Quantum ESPRESSO). Proficient in Python-based automation, data extraction and high-throughput workflows. Strong interest in generative models, ML-guided materials design and energy-relevant applications such as hydrogen production, CO_2 conversion, low-dimensional semiconductors and alloyed nanomaterials. Bridging computational modeling with experimental synthesis and data-driven research.

TECHNICAL SKILLS

Computational Analytical Skills: Electronic structure calculations using DFT, spin-orbit coupling(SOC) calculations, thermoelectric property prediction via Boltzmann transport theory, band structure and DOS analysis, orbital overlap integrals, charge density mapping, catalytic reaction pathway modeling, pressure-dependent material behavior, molecular dynamics simulations, crystal structure optimization, doping simulations in periodic systems, materials screening under data-scarce regimes and integration of quantum chemical results with physical property models.

Technical Proficiency: Skilled in VASP, Gaussian, BoltzTraP2, Quantum ESPRESSO (cp.x, pw.x, projwfc.x, and bands.x), VESTA, XCrySDen and Python-based libraries (NumPy, Matplotlib); scripting with Python and Bash for job automation and post-processing; working knowledge of Linux-based HPC environments and shell automation; familiar with Git version control and Jupyter Notebooks for reproducible research.

Data Visualization: Proficient in VESTA, XCrySDen, Origin and Matplotlib for visualizing molecular orbitals, charge densities, band structures, MD trajectories and structure-property relationships.

Experimental Skills: SEM, TEM, XRD, UV-Vis, IR, NMR, ESR and Mass Spectrometry.

EXPERIENCE

Indian Institute of Technology, Delhi, India | Project Fellow Jul 2025 - Dec 2025

- ***"Probing the Structure and Electronic Properties of Novel Two-Dimensional Au-Ag Alloys" (Mentor: Prof. N. Ray)***
  - Performed vc-relax, SCF and NSCF calculations to optimize structure and obtain accurate electronic states of 2D Au-Ag monolayers (HCP phase).
  - Conducted ab-initio molecular dynamics simulations at 300 K and 1500 K to evaluate thermal stability and atomic-scale dynamics.

  Tools & Techniques: Quantum ESPRESSO (cp.x), XCrySDen, Bash
- ***"Interface-Driven Modulation in CNT-Encapsulated and 1D Halide Perovskites" (Mentor: Prof. N. Ray)***
  - Performed DFT simulations of Cs_4PbI_5 encapsulated in carbon nanotubes with B/N doping to analyze stability and electronic modifications.
  - Conducted spin-orbit coupling (SOC) calculations on $CsPbI_3$ and Cs_4PbI_5 nanowires to study relativistic effects on their electronic structures.

  Tools & Techniques: Quantum ESPRESSO (pw.x, projwfc.x, bands.x), XCrySDen, Bash

Ashoka University, Sonipat, India | DST Research Fellow Oct 2024 - Dec 2024

- ***"Computational design and development of transition metal pincer complexes for dehydrogenation of methanol and formic acid to produce hydrogen" (Mentor: Prof. V. Avasare)***
  - Conducted DFT-based mechanistic studies on hydrogen production via methanol and formic acid dehydrogenation using Fe-based transition metal pincer complexes.
  - Evaluated the thermodynamic stability and electronic properties of proposed catalysts using Gaussian and VASP.

  Tools & Techniques: Gaussian, Origin

Indian Institute of Technology Roorkee, India | Master's Thesis Jan 2023 - May 2023

- ***"Computational study on the structure, electronic and thermoelectric properties of CoTiSb" (Mentor: Prof. H. C. Kandpal)***
  - Investigated the electronic structure and thermoelectric behavior of half-Heusler CoTiSb using density functional theory(VASP) in combination with Boltzmann transport theory (BoltzTraP2).
  - Examined the influence of external pressure, doping and temperature variation on carrier transport characteristics and optimization of the figure of merit (zT).

  Tools & Techniques: VASP, BoltzTraP2, VESTA, Origin

Indian Institute of Science (IISc), Bangalore, India | Summer Research Intern Jun 2022 - Aug 2022

- ***"Computational determination of the stabilities of icosahedral arrangements of boranes by substituting one of the atoms to Si, Ge and Sn" (Mentor: Prof. E.D. Jemmis)***
  - Conducted DFT-based electronic structure calculations to evaluate the relative stabilities of substituted icosahedral borane clusters upon site-specific replacement with Si, Ge and Sn atoms.
  - Employed overlap integral analysis to quantify bonding interactions between ring and capping atoms, providing insight into the electronic delocalization and structural robustness of the modified clusters.

  Tools & Techniques: Gaussian, Origin, Python, Shell Programming

EDUCATION

Indian Institute of Technology Roorkee, India Jul 2021- May 2023
Masters in Chemistry
Relevant Coursework: Quantum Chemistry, Symmetry and Group Theory, Thermodynamics, Interfaces and Solids, Kinetics and Photochemistry, Molecular Spectroscopy, Nanoscale Materials - Properties and Applications, Structure and Reactivity of Organic Molecules, Advanced Coordination Chemistry, Advanced Organometallic Chemistry, Supramolecular Chemistry, Organic Structure Determination, Inorganic Biochemistry and Reaction Mechanism, Frontiers in Bioinorganic Chemistry.

University of Delhi, India Jul 2018 - May 2021
Bachelors with Honours in Chemistry
Relevant Coursework: Quantum Chemistry, Phase Equilibria and Electrochemical Cells, Conductance and Chemical Kinetics, Coordination Chemistry, Heterocyclic Chemistry, Intellectual Property Rights, Biomolecules, Industrial Materials, Organometallic Chemistry, Analytical Chemistry

PUBLICATIONS

- M. Godara, S. Shome, S.S. Birwal, N. Ray, *Interface-Driven Modulation of Structure and Electronic Properties in CNT-Encapsulated $CsPbI_3$ and Cs_4PbI_5 1D Perovskites* [Accepted]
- K. Yadav, S. Shome, S.S. Birwal, N. Ray, *Probing the Structure and Electronic Properties of Novel Two-Dimensional Au-Ag Alloys* [Manuscript under preparation]

TEACHING EXPERIENCES

- Chemistry: Teaching medical and engineering aspirants — June 2024 - Present
- Chemistry: FIITJEE Limited, India — May 2023 - May 2024
- Physics, Chemistry and Mathematics: Private Tutor — Apr 2019 - Jun 2022

CERTIFICATE COURSES

- Database Operations in MariaDB using Python — Infosys, Oct 2022
- Basic Image Classification with TensorFlow — Coursera, Oct 2022
- Foundation: Data, data everywhere — Google, Sep 2022
- Linear regression with NumPy and Python — Coursera, Sep 2022
- Satellite Photogrammetry and its application — ISRO, Jun 2021

WORKSHOPS/CONFERENCES

- Workshop on Spintronics, Electronics and Communication Engineering — IIT Roorkee, Dec 2024
- International Conference on Main Group Molecules to Materials — IIT Roorkee, Dec 2021
- Seminar on Advances in Polymer Nanocomposites — American Chemical Society, Oct 2021
- International Conference on Recent Trends in Drug Discovery and Development — University of Delhi, Oct 2021

ACHIEVEMENTS, AWARDS AND SCHOLARSHIPS

- IELTS - 7 Band Score — Feb 2025
- Dr. Ambedkar National Merit Award Scheme — Government of India, July 2018
- National Scholarships - Merit cum Means — University of Delhi, Mar 2019-2021
- Verbal Mention: UNEP — Model United Nations of IITR, Sep 2021
- Airman Selection — Indian Air Force, Aug 2020
- Third State Rank — International Olympiad of Science, Mar 2018
- Merit Certificate: PULSE — AIIMS Delhi, Sep 2019

POSITIONS OF RESPONSIBILITY

- Placement and Internship Coordinator — IIT Roorkee, May 2022 - May 2023
- Vocalist: Rhapsody, The Music Society of DDUC — University of Delhi, Aug 2019 - Oct 2019
- Folk Dancer: Kings 6 Bhangra, The Bhangra Club of DDUC — University of Delhi, Nov 2018 - Mar 2021
- Core Member: Vyaktitav, The Personality Development Cell of DDUC — University of Delhi, Nov 2018 - Nov 2019
- Lab Coordinator: Responsible for lab safety and order of chemicals — University of Delhi, Oct 2018 - Nov 2019
- Vice President: Escapade, The Adventure Club of DDUC — University of Delhi, Sep 2018 - Mar 2021

OTHERS

- Extra-Curricular Activities: Member of Dance and Music society at University of Delhi
- Hobbies: Guitar, Pencil Sketching, Vector and Graphic Designing, Video Editing and Mixing
- Language: English, Hindi, Punjabi

REFERENCES (AVAILABLE UPON REQUEST)

- Prof. Eluvathingal D. Jemmis, Emeritus Professor — IISc Bangalore, India; jemmis@iisc.ac.in
- Prof. Nirat Ray, Associate Professor — IIT Delhi, India; nirat@iitd.ac.in
- Prof. Hem C. Kandpal, Assistant Professor — IIT Roorkee, India; hem.kandpal@iitr.ac.in